MSD MANUAL
Consumer Version

COVID-19

(Coronavirus Disease 2019; COVID)

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COVID-19 is an acute respiratory illness that can be mild or severe and is caused by a coronavirus named SARS-CoV-2.

- Symptoms of COVID-19 vary significantly.
- Two types of tests can be used to diagnose a COVID-19 infection.
- Measures should be taken to prevent infection, particularly vaccination.
- Treatment of COVID-19 depends on how severe the illness is and the likelihood that the person will develop severe disease.

COVID-19 Virus

3D MODEL

COVID-19 was first reported in late 2019 and has since spread extensively worldwide. For current information on the number of cases and deaths, see the Centers for Disease Control and Prevention: COVID Data Tracker and the World Health Organization Coronavirus (COVID-19) Dashboard.

The coronavirus that causes COVID-19 is called SARS-CoV-2. It is related to but slightly different from the coronavirus that causes SARS.

Social determinants of health (conditions in the places where people are born, live, learn, work, and play) impact a wide range of health risks and outcomes, such as exposure to SARS-CoV-2 infection, severe COVID-19, and death, as well as access to testing, vaccination, and treatment. In the United

States, COVID-19 case, hospitalization and death rates were higher during the height of the pandemic in some racial and ethnic minority groups, including among people who are Black, Hispanic or Latino, American Indian, and Alaska Native.

Most viruses, including the SARS-CoV-2 virus, evolve and new variants of the virus emerge. Variants with the potential to be more transmissible, more likely to cause severe disease, or that are harder to treat or detect are tracked as Variants of Concern and are commonly referred to by their Greek alphabet label. The dominant variants in the United States and much of the world are Alpha, Beta, Delta, and Omicron. The Omicron variant has been the most common variant worldwide since March 2022, with newer and more transmissible Omicron subvariants replacing the original Omicron.

Transmission of COVID-19

COVID-19 is mainly spread from person to person through respiratory droplets that are produced when an infected person coughs, sneezes, sings, exercises, or talks. The virus spreads through both large respiratory droplets that can travel short distances and through small respiratory particle aerosols that can linger in the air for several hours and travel longer distances (up to about 6 feet [2 meters]) before being inhaled.

In general, the closer and longer the interaction with an infected person, the higher the risk of the virus spreading. Factors such as distance from an infected person, duration of time in the presence of an infected person, the size of the air space, and the direction and speed of airflow can contribute to this risk. People may also get a COVID-19 infection by touching something that has the virus on it and then touching their own mouth, nose, or eyes.

The virus is usually transmitted by a person with symptoms of the infection. However, the virus can be transmitted by people before they exhibit symptoms (presymptomatic) and even by people who are infected but never develop symptoms (asymptomatic).

Situations with high risk of transmission include congregate living facilities (for example, elder care or other long-term care facilities, residential schools, prisons, ships) and crowded, poorly ventilated environments (such as indoor religious services, gyms, bars, night clubs, indoor restaurants, and meat-packing facilities). The residents of elder care facilities are also at high risk of severe disease because of age and underlying health conditions.

Large indoor events, such as meetings or weddings, have also been associated with high transmission rates. These have been called super-spreader events and may lead to high rates of infection due to large numbers of people being physically close to each other without adequate airflow or ventilation.

Viral tests in people with COVID-19 may remain positive for at least 3 months regardless of symptoms. However, even people with lingering symptoms are generally not considered infectious after 10 days of illness.

Symptoms of COVID-19

Symptoms vary in people with COVID-19 and may include the following:

- Fever
- Cough
- Sore throat
- Congestion or runny nose
- Shortness of breath or difficulty breathing
- Chills or repeated shaking with chills
- New loss of smell or taste
- Fatigue
- Muscle pain
- Headache
- Nausea or vomiting
- Diarrhea

If people develop symptoms, they usually appear about 2 to 10 days (incubation period) after being infected. For the Omicron variant, symptoms usually appear in only 2 to 4 days. Many infected people have no symptoms or have only mild disease.

The risk of serious disease and death in people with COVID-19 increases

- With age
- In people who smoke or previously smoked cigarettes
- In people with other serious medical disorders, such as cancer or chronic heart, lung, kidney, or liver disease, sickle cell disease, cystic fibrosis, diabetes, obesity, or immunocompromising disorders

For all age groups, risk of serious disease and death decreases in people who are vaccinated against COVID-19.

Complications

In addition to respiratory disease that can be severe and lead to death, other serious complications include

- Heart disorders including arrhythmias, heart muscle disorders, and acute heart injury
- Coagulation disorders including blood clots in small and large blood vessels as well as bleeding
- Guillain-Barré syndrome (rare)

A rare complication called multisystem inflammatory syndrome in children (MIS-C) that may be linked to COVID-19 has been reported in children. Symptoms of this can be similar to the rare condition Kawasaki disease and include fever, abdominal pain, and rash. Vaccination helps protect against developing MIS-

C. A similar complication has been reported in young and middle-aged adults (multisystem inflammatory syndrome in adults [MIS-A]).

Some people with COVID-19 develop Long COVID. Long COVID is defined as a chronic condition that occurs after SARS-CoV-2 infection and is present for at least 3 months. Long COVID includes a wide range of symptoms or conditions that may improve, worsen, or be ongoing. Long COVID occurs more often in people who had severe COVID-19 illness, but anyone who gets COVID-19 can experience it, including children (see also Centers for Disease Control and Prevention [CDC]: Long COVID).

Diagnosis of COVID-19

- Tests to identify the virus

Doctors suspect COVID-19 in people who have symptoms of the infection. Recent close contact with someone who has COVID-19 increases the likelihood of infection. People who suspect they may have COVID-19 should call their doctor before being tested and before arriving at a clinic so appropriate precautions can be taken.

The following people should be tested for COVID-19 (also see COVID-19 Testing):

- People with symptoms of COVID-19
- People who have been in close contact with a COVID-19–infected person but do not have symptoms—get tested at least 5 days after exposure; wearing a mask is recommended until negative test results are received
- People asked to test because of school, workplace, health care setting, or government requirements

People who took part in activities that put them at higher risk for COVID-19, such as attending large social gatherings or being in crowded indoor settings without correct and consistent masking, may also want to be tested.

There are 2 types of tests to diagnose COVID-19 infection:

- NAAT (nucleic acid amplification test)
- Antigen tests

There are multiple types of NAATs. RT-PCR (real-time reverse transcriptase–polymerase chain reaction) tests, often simply called PCR tests, are the type of NAAT that have the highest sensitivity and specificity, meaning they are more accurate and thus the preferred initial test for COVID-19. RT-PCR tests can be done on upper and lower respiratory secretions (samples from nasal or oral swabs or saliva) to identify the virus. Test instructions must be carefully followed.

Antigen tests can be done at home or in a health care setting, but they are generally less accurate than NAATs, including RT-PCR tests. (See also FDA: At-Home COVID-19 Antigen Tests-Take Steps to Reduce Your Risk of False Negative Results.) Therefore, it may be necessary to confirm some antigen test results (for example, a negative test in a person with symptoms) with a RT-PCR or other NAAT. Many antigen

test kits also recommend repeating the test over several days to increase the likelihood of detecting infection. Also, some tests may not detect the Omicron variant or other emerging variants (see FDA: SARS-CoV-2 Viral Mutations: Impact on COVID-19 Tests).

Another type of test is called an antibody test. Antibody tests (also called serologic tests) are not used for diagnosing current infections. Antibody tests help determine whether the person being tested was previously infected, which is important for tracking cases and studying the virus.

Treatment of COVID-19

- Medications to relieve fever and muscle aches
- Sometimes, for people with mild to moderate illness with high-risk of severe disease: nirmatrelvir/ritonavir combination; remdesivir (short course); molnupiravir
- For severe illness: remdesivir; dexamethasone; immunomodulators

Treatment of COVID-19 depends on how severe the illness is and the likelihood that the person will develop severe disease.

For mild illness, resting at home is often sufficient. Acetaminophen or a nonsteroidal anti-inflammatory drug (NSAID), such as ibuprofen, may be taken to relieve fever and muscle aches. Despite initial anecdotal concerns, there is no scientific evidence that the use of NSAIDs worsens COVID-19. Similarly, there is no scientific evidence that people with COVID-19 who take the blood pressure medications called angiotensin-converting enzyme inhibitors (ACEI) or angiotensin receptor blockers (ARBs) should stop taking them.

For more severe illness, some people require hospitalization with treatment that might include supplemental oxygen or mechanical ventilation.

For people with severe illness from COVID-19 or people at high risk for progression to severe disease, some medications and other therapies are recommended. For example, some immunomodulator therapies (such as tocilizumab, baricitinib, or sarilumab) may be used for severe disease. This is a rapidly evolving topic (see National Institutes of Health (NIH) COVID-19 Treatment Guidelines and Infectious Diseases Society of America (IDSA) Guidelines on the Treatment and Management of Patients with COVID-19).

Nirmatrelvir/ ritonavir is a combination antiviral medication taken by mouth. It may be used to treat mild to moderate COVID-19 infection in some adults and adolescents who are at high risk for progression to severe COVID-19, including hospitalization or death. It is not authorized for use for longer than 5 consecutive days.

Some people have symptoms return after using nirmatrelvir/ritonavir, and tests for SARS-CoV-2 can become positive again, even in people who do not have symptoms return. Additional treatment is not currently recommended, but people should isolate if rebound symptoms or positive tests occur. (See also CDC: COVID-19 Rebound After Paxlovid Treatment.)

Remdesivir (an antiviral medication) is for the treatment for select people with COVID-19. Remdesivir is given intravenously. The recommended treatment duration is 3 to 10 days. The combination of remdesivir and the corticosteroid **dexamethasone** is commonly used in hospitalized people who need supplemental oxygen.

Molnupiravir is an antiviral medication taken by mouth. It may be used to treat mild to moderate COVID-19 infection in nonhospitalized adults who are at high risk for progression to severe COVID-19, including hospitalization or death, and for whom alternative COVID-19 treatment options are not available or appropriate. Molnupiravir is not authorized for use for longer than 5 consecutive days. It is not recommended for use during pregnancy.

Bamlanivimab/etesevimab, casirivimab/imdevimab, sotrovimab, and bebtelovimab are monoclonal antibody therapies. They are not effective against the Omicron variant. Therefore, the FDA recommended against their use in treatment of COVID-19, because Omicron has become the dominant variant in the United States.

The following therapies are also NOT recommended for the treatment or prevention of COVID-19:

- Vilobelimab, a monoclonal antibody (not yet recommended in clinical guidelines)
- Blood plasma from recovered patients
- Nonspecific immunoglobulin (IVIG) and mesenchymal stem cell therapy
- Additional immunomodulatory therapies, including interferons, kinase inhibitors, and interleukin inhibitors
- Azithromycin and antiretrovirals
- Lopinavir/ritonavir (an HIV retroviral)
- Chloroquine and hydroxychloroquine (antimalaria medications)
- Ivermectin (an antiparasite medication): the FDA and other organizations have issued warnings about toxicity from the inappropriate use of ivermectin preparations intended for large animal use (see FDA: Ivermectin and COVID-19).

Prevention of COVID-19

Vaccination

The best way to prevent infection, severe illness, and death from COVID-19 is to be up to date with COVID-19 vaccinations. Unvaccinated people are more likely to die from COVID-19 than vaccinated people.

For a detailed discussion of the vaccines approved in the United States, see COVID-19 Vaccine.

Multiple COVID-19 vaccines are currently in use worldwide. For more information on global vaccine approvals and clinical trials, see the World Health Organization's COVID-19 vaccine tracker and landscape.

Following the recommendations for vaccination is very important for people's own health and for the health of their family and the people in their community.

The CDC recommends that people aged 6 months and older receive the 2023–2024 updated COVID-19 vaccine to protect against serious illness from COVID-19. The CDC also recommends adults ages 65 years and over receive an additional updated 2023-2024 COVID-19 vaccine dose (see CDC: Older Adults Now Able to Receive Additional Dose of Updated COVID-19 Vaccine).The vaccines are manufactured by Pfizer-BioNTech, Moderna, or Novavax. The updated 2023–2024 vaccines target the XBB.1.5 Omicron variant and replace the previously used vaccinations. See CDC: Stay Up to Date with COVID-19 Vaccines for details regarding manufacturers and number of doses recommended depending on age, prior vaccination history, and immunocompromised status.

Masking and other routine measures

In addition to staying up to date with COVID-19 vaccines, people can avoid being exposed to the virus by washing hands frequently, wearing face masks, maintaining a good social distance from other people, avoiding poorly ventilated spaces and crowds, and taking other steps recommended by the Centers for Disease Control and Prevention (CDC).

The CDC recommends the following people wear a well-fitting face mask, covering both mouth and nose (see CDC: Masks and Respiratory Viruses Prevention):

- People in communities where respiratory viruses are causing a lot of illness.
- People who have been exposed or are around people who were recently exposed to a respiratory virus, are sick, or are recovering.
- People or others around them have risk factors for severe illness.

In addition to following the CDC recommendations, people may be required to wear a mask by local laws, regulations, or rules or business or workplace guidance, and this may vary by vaccination status. People who are at increased risk for severe disease or who have someone in their household at increased risk might choose to wear a mask regardless of any requirements. People who are at increased risk for severe disease include those who are unvaccinated, have a weakened immune system, have an underlying medical condition, are pregnant or recently pregnant, and people older than 65 (see Risk Factors for Severe Illness from Respiratory Viruses). Different types of masks provide different levels of protection, including (in increasing order of protection): multi-layer cloth masks; multi-layer surgical masks and K95 masks; and N95 masks (see CDC: Community Respirators and Masks).

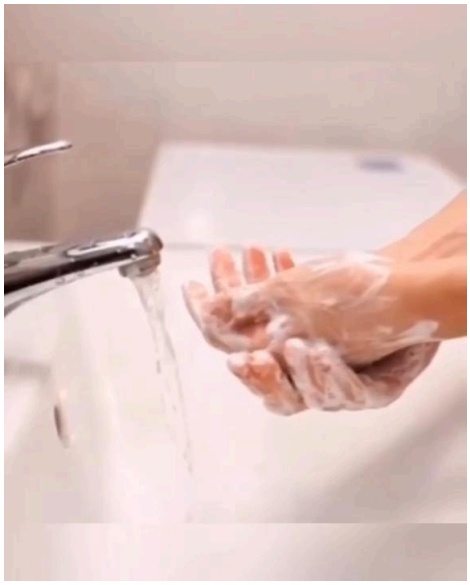
In addition to being up to date with vaccinations and wearing a mask, the following measures help prevent the spread of COVID-19 and other respiratory viruses:

- People with symptoms of COVID-19 should consider testing (see CDC: Testing and Respiratory Viruses)
- If at increased risk of getting very sick from COVID-19, avoid crowded places and indoor spaces that do not have fresh air from the outdoors (see CDC: Taking Steps for Cleaner Air for Respiratory Virus Prevention)

- Maintain good social distance from other people, especially if at higher risk of getting very sick with COVID-19 (see CDC: About Physical Distancing and Respiratory Viruses)
- If possible, maintain good social distance between a person who is sick with COVID-19 and other household members
- Wash hands often with soap and water for at least 20 seconds, especially after going to the bathroom, before eating, and after blowing the nose, coughing, or sneezing (see CDC: Hygiene and Respiratory Viruses Prevention)
- Use an alcohol-based hand sanitizer with at least 60% alcohol if soap and water are not readily available
- Avoid touching eyes, nose, and mouth with unwashed hands
- Stay home when sick (see CDC: Preventing Spread of Respiratory Viruses When You're Sick)
- Cover cough or sneeze with a tissue, then throw the tissue in the trash
- Clean and disinfect frequently touched objects and surfaces using a regular household cleaning spray or wipe
- Monitor health for potential symptoms and take temperature if symptoms develop

How To Wash Your Hands

VIDEO



Measures for those exposed or infected

People who do not have symptoms but have been **exposed** (in close contact) to a COVID-19–infected person should consider getting tested at least 5 days after exposure (or sooner if they develop symptoms). Exposed people should wear a mask when around others until negative test results are received. See CDC: Testing and Respiratory Viruses for additional information regarding masking and testing after exposure.

For people **infected** with COVID-19, isolation and precaution measures are recommended in an attempt to limit the spread of SARS-CoV-2 infection (see CDC: Preventing Spread of Respiratory Viruses When

You're Sick). People with symptoms should stay home and separated from others. People can return to their normal activities after 24 hours without a fever (without using fever-reducing medication) and with other symptoms improving. For an additional 5 days after symptoms improve, they should wear a mask when around others both in public and at home and take other precautions.

More Information

The following English-language resources may be useful. Please note that THE MANUAL is not responsible for the content of these resources.

CDC: Vaccines for COVID-19
CDC: Respiratory Virus Guidance

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